

Advanced challenge (Due: 2024, Feb 1st)

Instructor: Sungshin Kim (Hanyang University)

Using the hierarchical bandit task environment (Figure 1) explained in the video instruction, implement a model-free Q-learning agent and a hybrid (model-free and model-based) Q-learning agent. For the hybrid agent, you can consider the parameters shown in Supplementary Table 1 (see the reference), or you can tune them to learn the task.

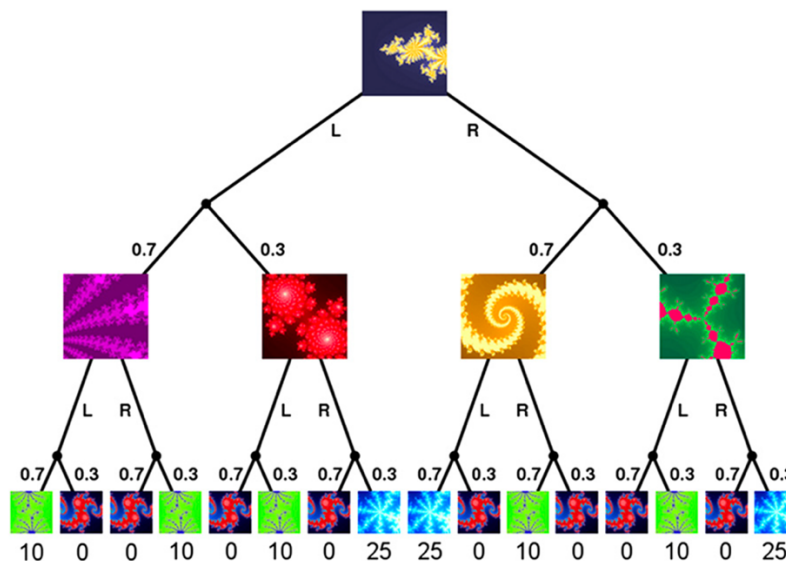


Figure 1. Structure of hierarchical decision-making task (Adapted from the reference)

- (1) Plot and compare the learning curves of two agents from the simulation of multiple experiments (mean and their standard error of the mean).
- (2) Visualize how each of the two agents learned all the state-action value functions, $Q(s,a)$ across trials and report the final learned values. Then, confirm that the two agents learned the optimal state-action values as shown in Supplementary Figure 1 (see the reference).

- (3) Explain how the optimal state-action values shown in Supplementary Figure 1 are calculated.
- (4) For the hybrid agent, visualize the hybrid agent learned transition probabilities as shown in Supplementary Figure 2 (see the reference) across trials. Then, confirm that the agent learned all the transition probabilities, as designed in the environment.

Reference

Gläscher, J., Daw, N., Dayan, P., & O'Doherty, J. P. (2010). States versus rewards: dissociable neural prediction error signals underlying model-based and model-free reinforcement learning. *Neuron*, 66(4), 585-595.